

Tutor answers & hints for students

Tactic Hints

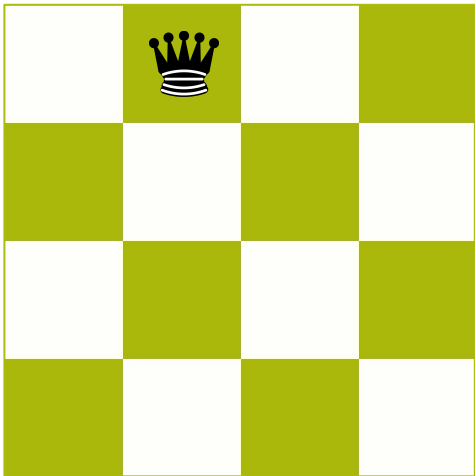
For some grids it's a lot harder for students if they start with a corner, so sometimes avoiding the corners early on can be helpful

The same with going straight to the center, it cuts off a lot without being guaranteed to be actually under any answer

Add more detail, more specific and more clear hints from the tutors to the students

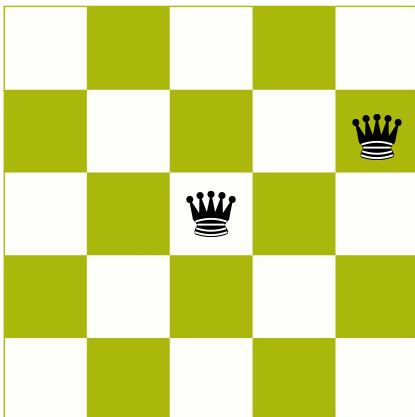
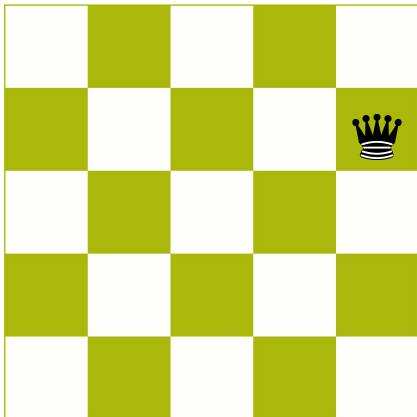
4 Queens

Here there are 2 total and 1 unique solution, if students are struggling give them the following confirmed queen:



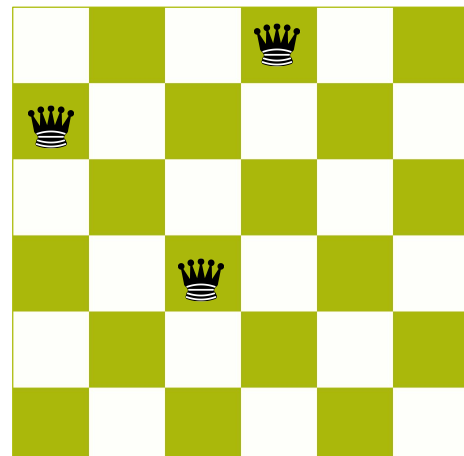
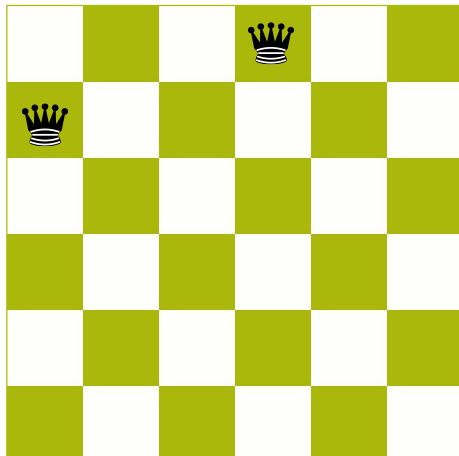
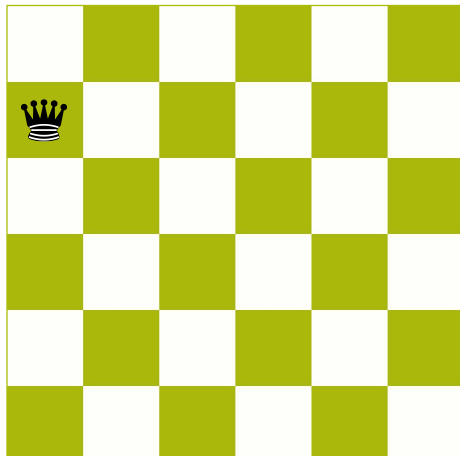
5 Queens

Here there are 10 total and 2 unique solution, if students are struggling you can give them the following confirmed queen, adding another if they continue to have trouble



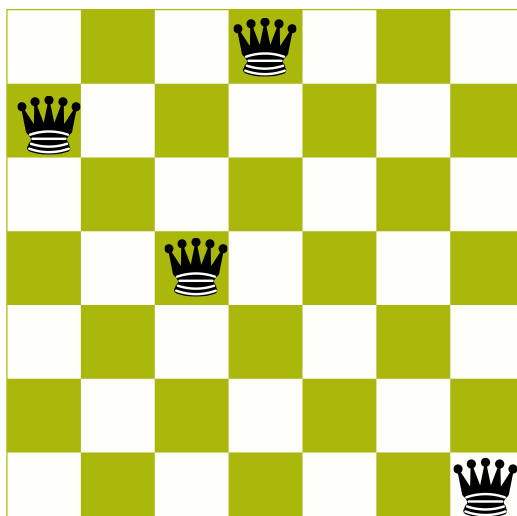
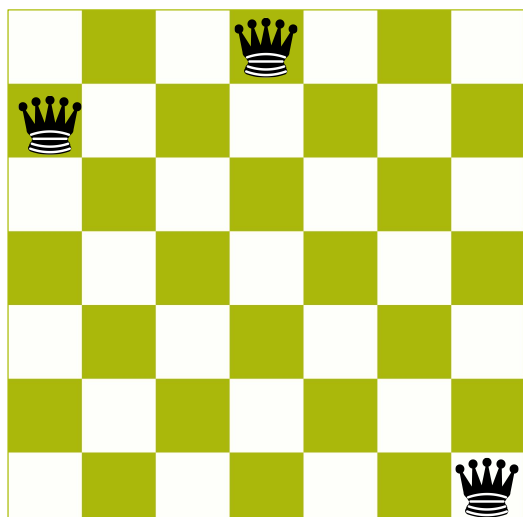
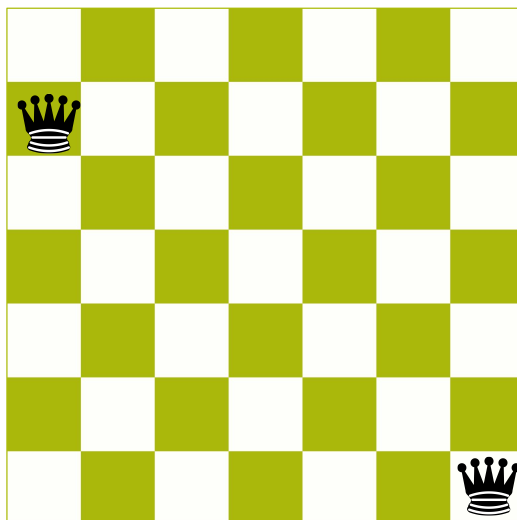
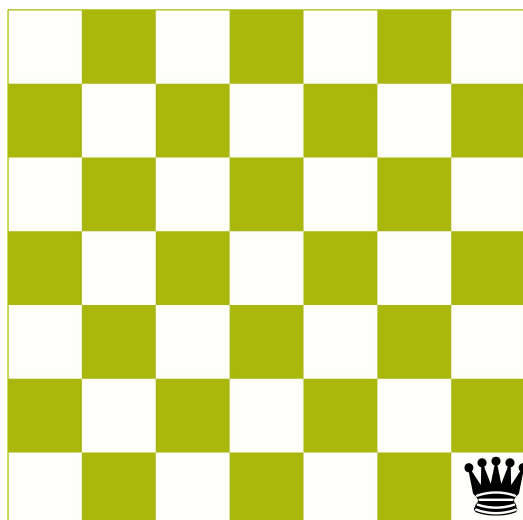
6 Queens

Here there are 4 total and 1 unique solution, if students are struggling you can give them the following confirmed queen, adding another if they continue to have trouble



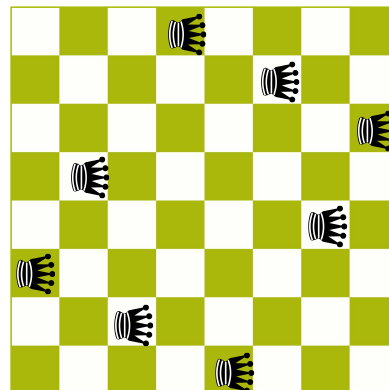
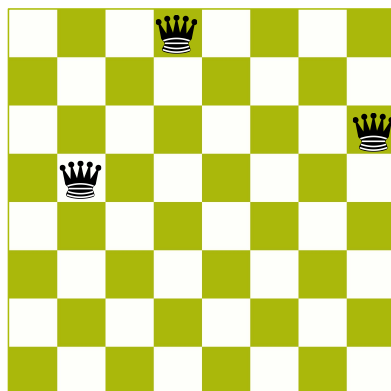
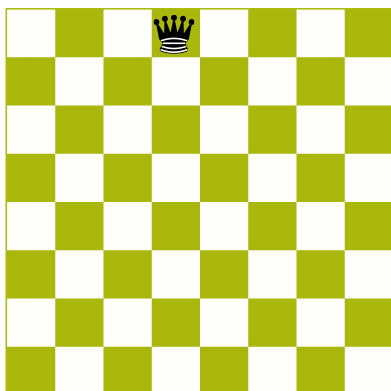
7 Queens

Here there are 40 total and 6 unique solution, if students are struggling you can give them the following confirmed queen, adding another if they continue to have trouble

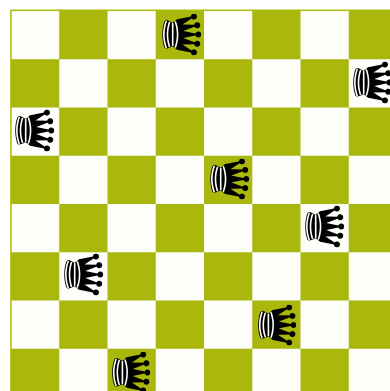
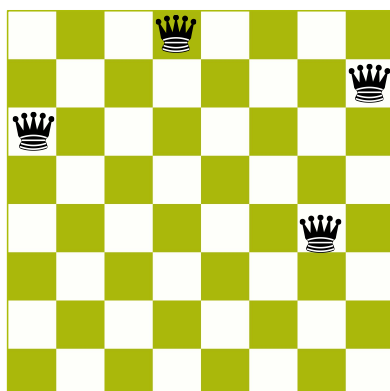
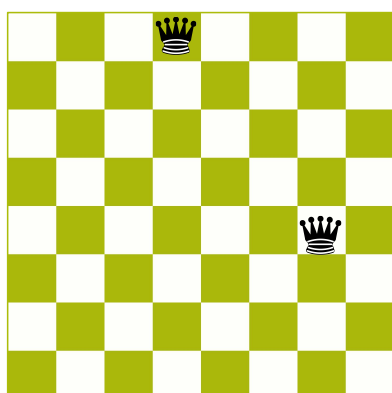
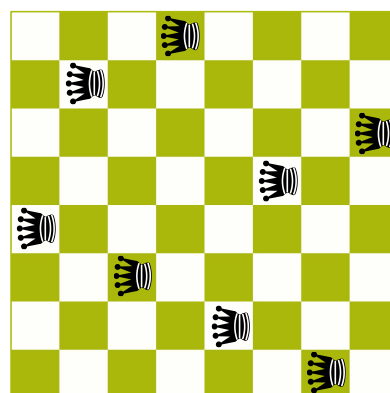
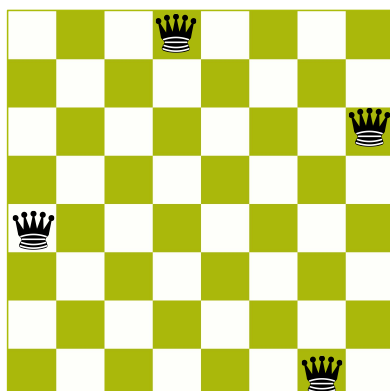
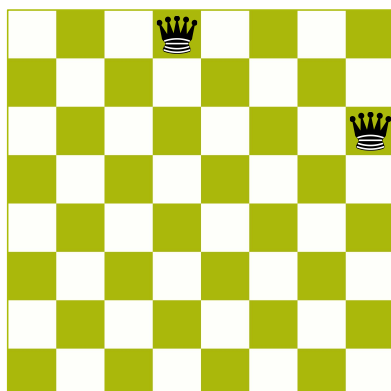


8 Queens

Here there are 92 total and 12 unique solutions, if students are struggling you can give them the following confirmed queen, adding another if they continue to have trouble



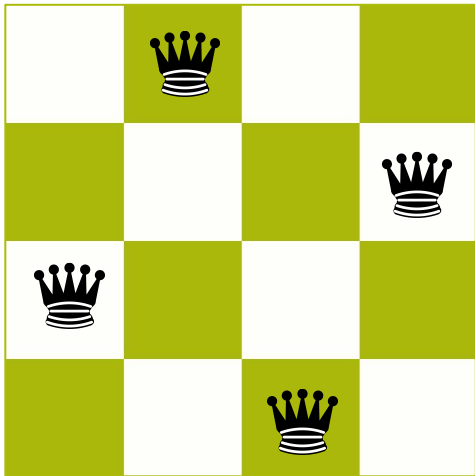
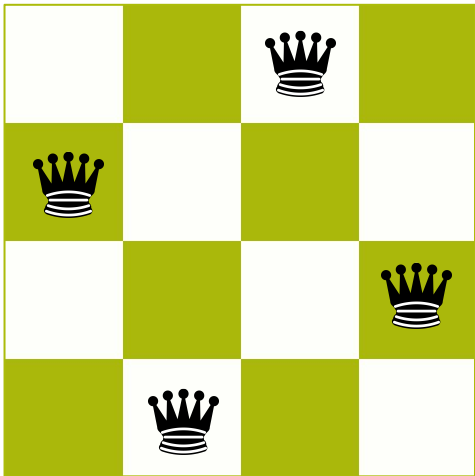
If they are working on the bonus and already have the above unique version try giving them these queens:



answers; tutors can
just check manually,
don't need to be in
depth

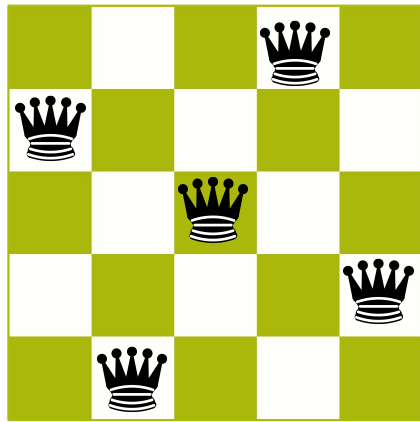
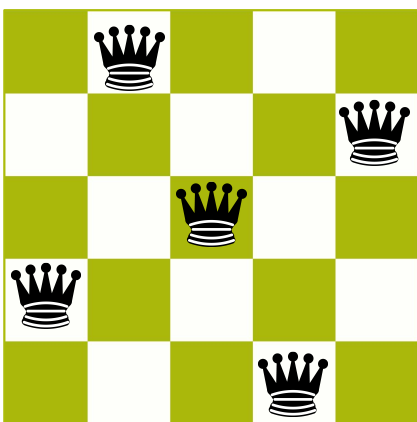
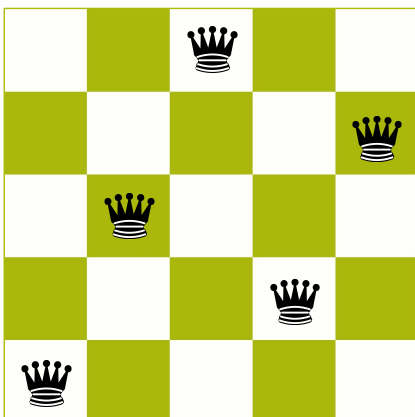
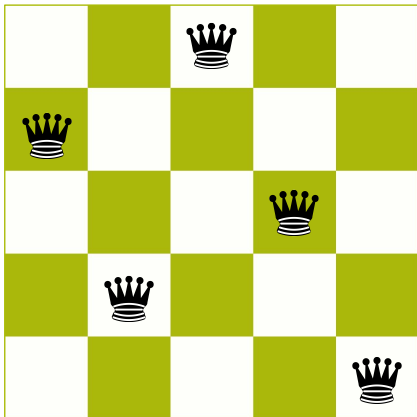
4 Queens

Here are all the solutions (2 total, 1 unique)



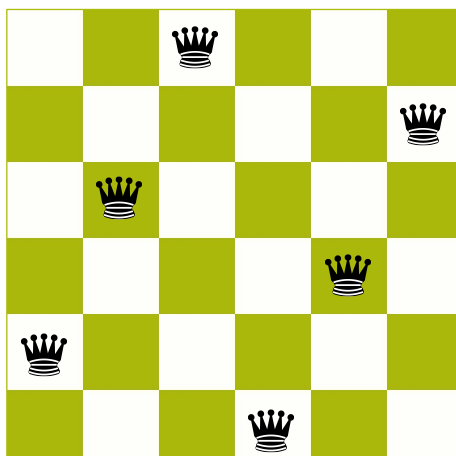
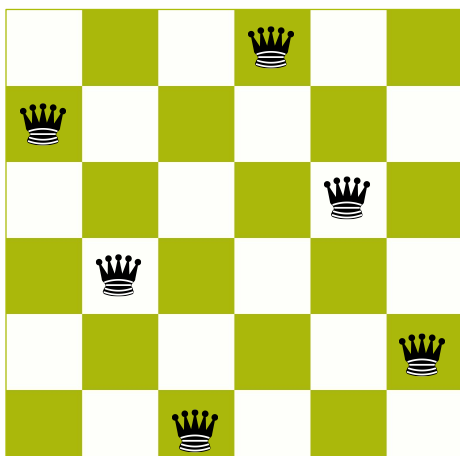
5 Queens

Here are all the unique solutions (10 total, 2 unique), flipped horizontally and vertically, so rotate the solution until it matches the students versions



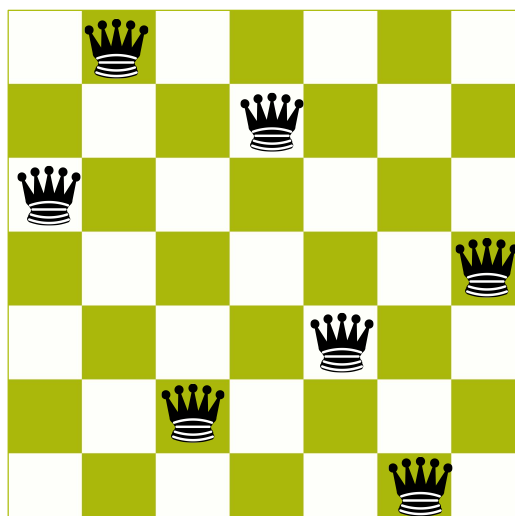
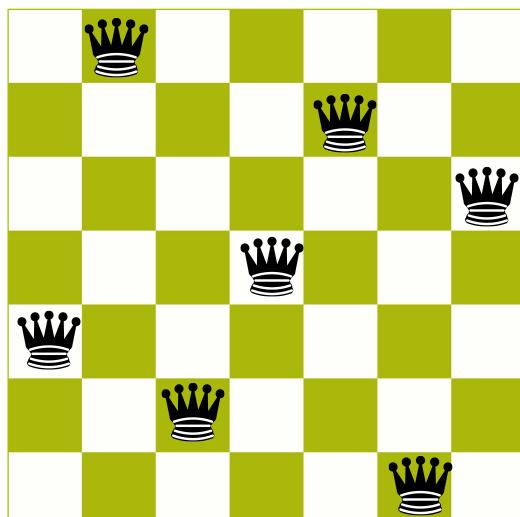
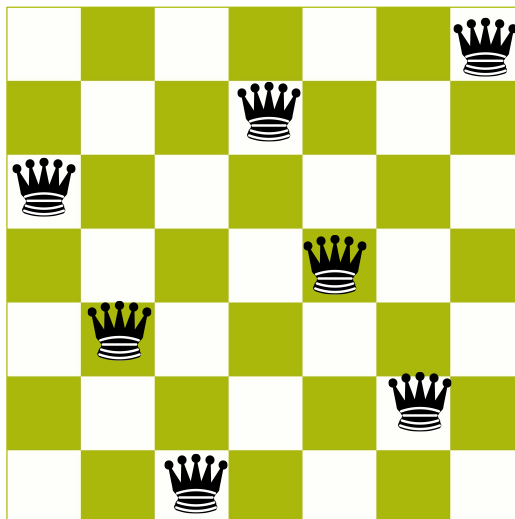
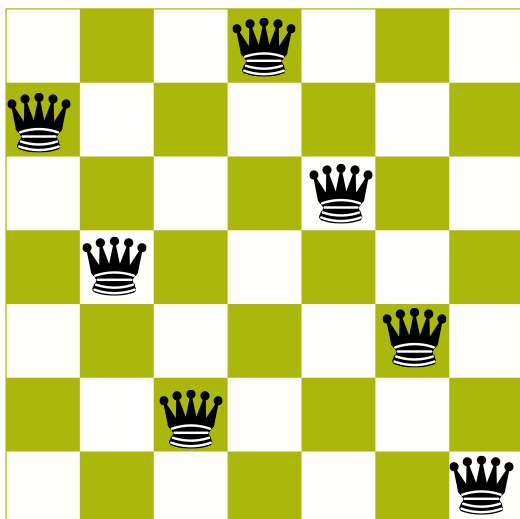
6 Queens

Here are all the unique solutions (4 total, 1 unique), flipped horizontally and vertically, so rotate the solution until it matches the students versions



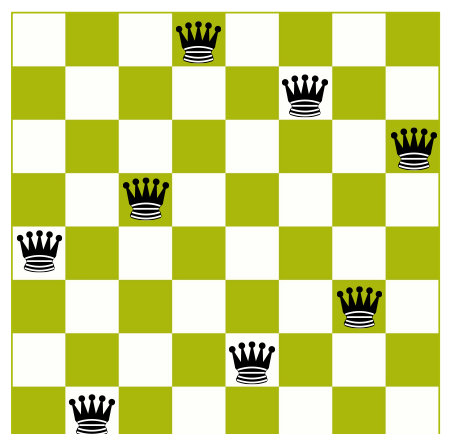
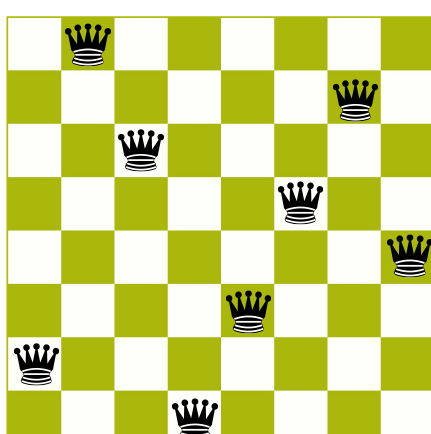
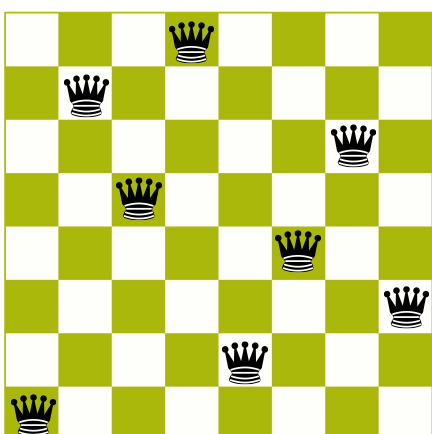
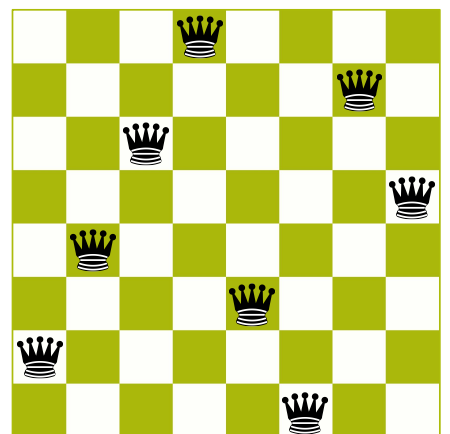
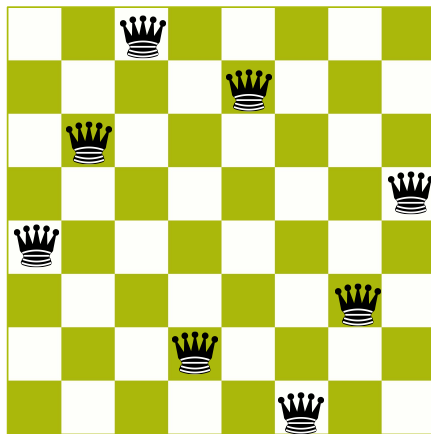
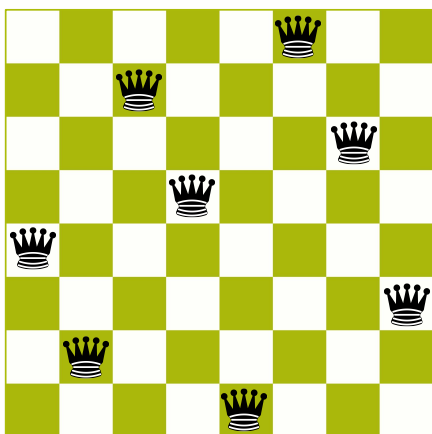
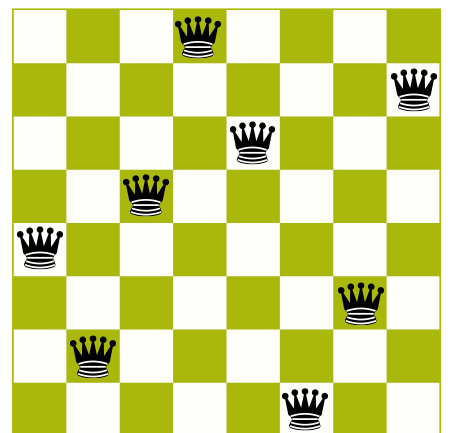
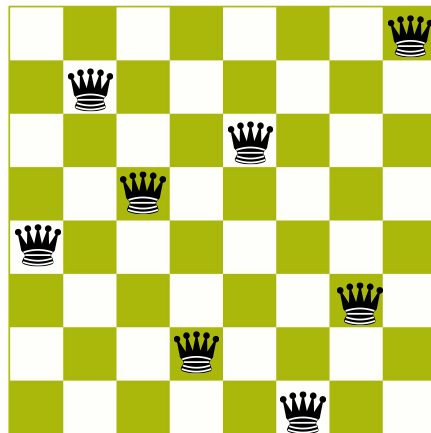
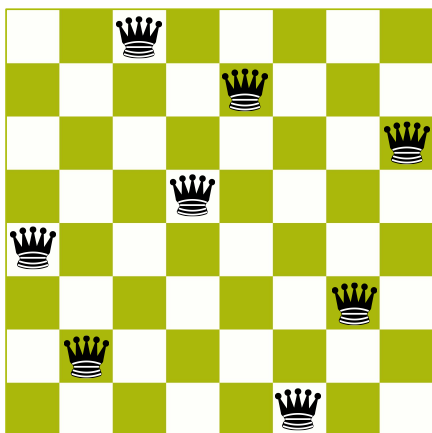
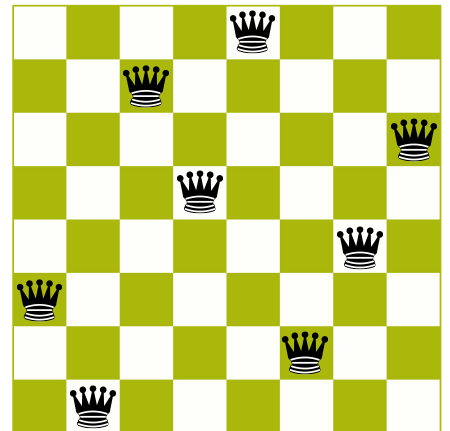
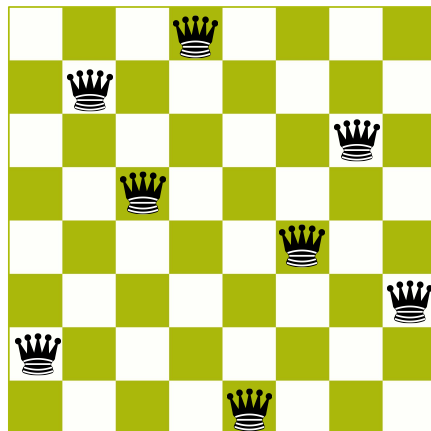
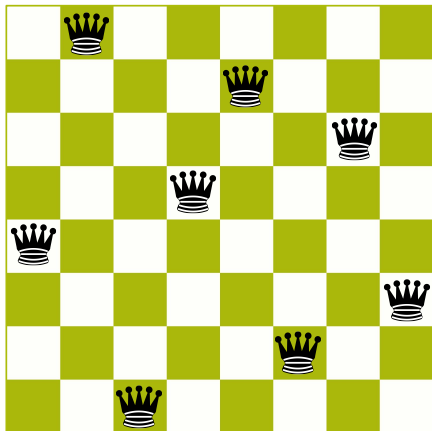
7 Queens

Here are all the unique solutions (40 total, 6 unique) although only 4 have been drawn out here



8 Queens

Here are all the unique solutions (92 total, 12 unique)



Reading the solutions

So you have to check a students answers... here are some hot tips on how to!

1. Start by checking each row & column each only have one queen!
(e.g. the coordinates should only have one a, or one 1)
2. Let's look at the set of numbers we have on our answers!

4,8,1,5,7,2,6,3

These numbers mean that a queen will fit into the 4th square of the 1st row!

But, because of isomorphism we might need to rotate our thinking, so a queen might fit in the 4th square of the 1st column!

If you want to consider the grid inverted you'll have to read back to front!

Let's apply it!

A team comes to you with the following coordinate set

a3, b6, c2, d7, e5, f1, g8, h4

First check!

Each letter and number only appears once!

Next! Pick a solution set

4,8,1,5,7,2,6,3



Bonuses!

8 rooks

14 bishops

32 knights $\rightarrow n^2/2$